

Assignment Submission

Application Build and Release Automation Lab

GLA Smart Online Course Registration System

Student Name: Ajay Kumar
Roll Number: 22515990001
Course / Program: B.Tech Computer Science and Engineering
Semester: 5th Semester
Subject: Application Build and Release Automation Lab

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Terminal screenshots are included as execution evidence.

Question 1 - Linux Environment and File Management

Project directory creation, folders, course_info.txt, pwd, ls/find and file content verification.

```
Last login: Wed Sep 2 09:15:06 on console
ajaykumar@ajays-MacBook-Air ~ % cd ~
mkdir Ajay-GLA-CourseRegistration
cd Ajay-GLA-CourseRegistration
mkdir src docs backup
touch course_info.txt
nano course_info.txt
mkdir: Ajay-GLA-CourseRegistration: File exists
mkdir: src: File exists
mkdir: docs: File exists
mkdir: backup: File exists
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % rm -r Ajay-GLA-CourseRegistration
rm: Ajay-GLA-CourseRegistration: No such file or directory
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % ls
backup      course_info.txt  docs          src
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cd ~
ajaykumar@ajays-MacBook-Air ~ % rm -r Ajay-GLA-CourseRegistration
override r--r--r-- ajaykumar/staff for Ajay-GLA-CourseRegistration/.git/objects/6a/3763633a4849dfce3f42b57238d91bbf8c30aa? y
override r--r--r-- ajaykumar/staff for Ajay-GLA-CourseRegistration/.git/objects/d8/876f294f09b22d7556ec59ca6ee2f5141c5d2? ^C
ajaykumar@ajays-MacBook-Air ~ % cd ~
ajaykumar@ajays-MacBook-Air ~ % mkdir Ajay-GLA-CourseRegistration
ajaykumar@ajays-MacBook-Air ~ % cd Ajay-GLA-CourseRegistration
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % mkdir src
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % mkdir docs
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % mkdir backup
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % touch course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % pwd
/Users/ajaykumar/Ajay-GLA-CourseRegistration
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % ls
backup      course_info.txt  docs          src
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % find .
.
./docs
./backup
./course_info.txt
./src
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat course_info.txt
Student Name: Ajay Kumar
Roll Number: 22615990001
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration %
```

Terminal output screenshot

Question 2 - Git Repository and Version Control

Git initialization, configuration, staging, commits, status, diff and repository history.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git init
Initialized empty Git repository in /Users/ajaykumar/Ajay-GLA-CourseRegistration/.git/
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch -M main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git config user.name "Ajay Kumar"
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git config user.email "ajay.kumar_cs25@gl.a.ac.in"
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git config user.name
Ajay Kumar
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git config user.email
ajay.kumar_cs25@gl.a.ac.in
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % touch src/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % touch docs/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % touch backup/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    backup/
    course_info.txt
    docs/
    src/

nothing added to commit but untracked files present (use "git add" to track)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add .
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   backup/.gitkeep
    new file:   course_info.txt
    new file:   docs/.gitkeep
    new file:   src/.gitkeep

ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Initialize course registration project"
[main (root-commit) 98ea732] Initialize course registration project
 4 files changed, 5 insertions(+)
 create mode 100644 backup/.gitkeep
 create mode 100644 course_info.txt
 create mode 100644 docs/.gitkeep
 create mode 100644 src/.gitkeep
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git diff
diff --git a/course_info.txt b/course_info.txt
index 696abde..3ef5f97 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -3,3 +3,5 @@ Roll Number: 22515998001
```

Execution evidence - Part 1

```
+++ b/course_info.txt
@@ -3,3 +3,5 @@ Roll Number: 22515998001
Subject Name: Application Build and Release Automation Lab
Project Name: Online Course Registration System
Semester: 5th Semester
+Course/Program Name: B.Tech Computer Science and Engineering
+Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Add program details and project description"
[main c8a4039] Add program details and project description
 1 file changed, 2 insertions(+)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline
c8a4039 (HEAD -> main) Add program details and project description
98ea732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git switch -c course-search
Switched to a new branch 'course-search'
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
course-search
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % touch src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat src/course_search.txt
Course Search Module

This module allows students to search for available university courses.

Search Options:
1. Course Name
2. Course Code
3. Department
4. Semester
5. Faculty Name

The module displays course details, available seats, and course credits.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch course-search
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    src/course_search.txt

nothing added to commit but untracked files present (use "git add" to track)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Develop course search module"
[course-search bbf2831] Develop course search module
 1 file changed, 12 insertions(+)
 create mode 100644 src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch
* course-search
  main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
course-search
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git switch main
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % ls src
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline --all --graph --decorate
* bbf2831 (course-search) Develop course search module
* c8a4039 (HEAD -> main) Add program details and project description
* 98ea732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % █
```

Execution evidence - Part 2

Question 3 - Branching and Feature Development

Creation of course-search branch, feature file development, commit and merge into main.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git merge course-search
Updating c8a4639..bbf2831
Fast-forward
 src/course_search.txt | 12 ++++++++
 1 file changed, 12 insertions(+)
 create mode 100644 src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % ls src
course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat src/course_search.txt
Course Search Module

This module allows students to search for available university courses.

Search Options:
1. Course Name
2. Course Code
3. Department
4. Semester
5. Faculty Name

The module displays course details, available seats, and course credits.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline --all --graph --decorate
* bbf2831 (HEAD -> main, course-search) Develop course search module
* c8a4639 Add program details and project description
* 98ea732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git switch -c course-update
Switched to a new branch 'course-update'
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
course-update
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git diff
diff --git a/course_info.txt b/course_info.txt
index 3ef6f07..6ad56c7 100644
--- a/course_info.txt
+++ b/course_info.txt
@@ -1,7 +1,7 @@
Student Name: Ajay Kumar
Roll Number: 22015990001
Subject Name: Application Build and Release Automation Lab
-Project Name: Online Course Registration System
+Project Name: Smart Online Course Registration System
Semester: 5th Semester
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Update project name on course-update branch"
[course-update 21b56ab] Update project name on course-update branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git switch main
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Update project name on main branch"
[main 0b5e59c] Update project name on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
```

Execution evidence - Part 1

```
+Project Name: Smart Online Course Registration System
Semester: 5th Semester
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Update project name on course-update branch"
[course-update 21b56ab] Update project name on course-update branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git switch main
Switched to branch 'main'
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Update project name on main branch"
[main 0b5e59c] Update project name on main branch
 1 file changed, 1 insertion(+), 1 deletion(-)
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git merge course-update
Auto-merging course_info.txt
CONFLICT (content): Merge conflict in course_info.txt
Automatic merge failed; fix conflicts and then commit the result.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main
You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   course_info.txt

no changes added to commit (use "git add" and/or "git commit -a")
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat course_info.txt
Student Name: Ajay Kumar
Roll Number: 22015990001
Subject Name: Application Build and Release Automation Lab
<<<<<< HEAD
Project Name: GLA Digital Course Registration System
=====
Project Name: Smart Online Course Registration System
>>>>>> course-update
Semester: 5th Semester
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % nano course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % grep -nE '<<<<<<|====|>>>>>>' course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git add course_info.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git commit -m "Resolve project name merge conflict"
[main d69d31c] Resolve project name merge conflict
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline --all --graph --decorate
* d69d31c (HEAD -> main) Resolve project name merge conflict
|
| * 21b56ab (course-update) Update project name on course-update branch
| * | 0b5e59c Update project name on main branch
| //
| * bbf2831 (course-search) Develop course search module
| * c8a4639 Add program details and project description
| * 98ea732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration %
```

Execution evidence - Part 2

Question 4 - Merge Conflict Creation and Resolution

Conflicting project-name changes on two branches and successful conflict resolution.

```
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % pwd
/Users/ajaykumar/Ajay-GLA-CourseRegistration
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % find . -not -path './.git*'
.
./docs
./docs/.gitkeep
./backup
./backup/.gitkeep
./course_info.txt
./src
./src/.gitkeep
./src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat course_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: GLA Smart Online Course Registration System
Semester: 5th Semester
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat src/course_search.txt
Course Search Module

This module allows students to search for available university courses.

Search Options:
1. Course Name
2. Course Code
3. Department
4. Semester
5. Faculty Name

The module displays course details, available seats, and course credits.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch
  course-search
  course-update
* main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline --all --graph --decorate
* d69d51c (HEAD -> main) Resolve project name merge conflict
|
| * 21b56ab (course-update) Update project name on course-update branch
| * 0b5e59c Update project name on main branch
|/
* bbf2831 (course-search) Develop course search module
* c8a4039 Add program details and project description
* 98ae732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % find . -not -path './.git*'
.
./docs
./docs/.gitkeep
./backup
./backup/.gitkeep
./course_info.txt
./src
./src/.gitkeep
```

Terminal output screenshot

Question 5 - Final Verification and Repository Status

Final branch/status checks, Git graph, project structure and file-content verification.

```
1. Course Name
2. Course Code
3. Department
4. Semester
5. Faculty Name

The module displays course details, available seats, and course credits.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch
course-search
course-update
* main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git branch --show-current
main
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git log --oneline --all --graph --decorate
* d69d31c (HEAD -> main) Resolve project name merge conflict
|
| * 21b56ab (course-update) Update project name on course-update branch
| * | 0b5e59c Update project name on main branch
| |
| | * bbf2831 (course-search) Develop course search module
| | * c8a4039 Add program details and project description
| | * 98ea732 Initialize course registration project
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % find . -not -path './.git*'
.
./docs
./docs/.gitkeep
./backup
./backup/.gitkeep
./course_info.txt
./src
./src/.gitkeep
./src/course_search.txt
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat course_info.txt
Student Name: Ajay Kumar
Roll Number: 22515990001
Subject Name: Application Build and Release Automation Lab
Project Name: GLA Smart Online Course Registration System
Semester: 5th Semester
Course/Program Name: B.Tech Computer Science and Engineering
Project Description: This system allows students to search, select, and register for available university courses online.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % cat src/course_search.txt
Course Search Module

This module allows students to search for available university courses.

Search Options:
1. Course Name
2. Course Code
3. Department
4. Semester
5. Faculty Name

The module displays course details, available seats, and course credits.
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % git status
On branch main
nothing to commit, working tree clean
ajaykumar@ajays-MacBook-Air Ajay-GLA-CourseRegistration % █
```

Terminal output screenshot